SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	**CSA1008**	**Course Title:**	**Software Engineering**
CO Assessed:	**CO2**	**Assessment:**	**Assessment Tool 2 – Scenario-Based Requirement Analysis**
Weightage:	**20%**	**Total Marks:**	**25**
CO2 Statement:	Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills		
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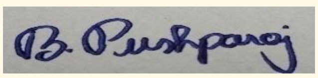
Date:	**06.8.2026**	**Duration:**	**60 minutes**

Code of Conduct

I B .Pushparaj(192421136) **certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.**

Signature of the Student: _ _____  ___________________

Annexure A – Scenario-Based Requirement Analysis

Instructions to Students:
Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

Question

Based on the assigned problem statement, perform a Scenario-Based Requirement Analysis and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

1. **Study and Analyze the Scenario**
    - Carefully understand the given problem statement.

- Identify the business domain, stakeholders, existing challenges, and system objectives.

2. **Identify Functional and Non-Functional Requirements**
    - List all functional requirements required by the proposed system.
    - Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability**.**
2. **Identify Actors and Use Cases**
    - Identify all primary and secondary actors interacting with the system.
    - List the major use cases associated with each actor.
    - Draw a Use Case Diagram representing the interactions between actors and the system.
2. **Prepare the Software Requirement Specification (SRS)**
   Develop an SRS document containing:
    - Introduction
    - Problem Description
    - Scope of the System
    - Functional Requirements
    - Non-Functional Requirements
    - Use Case Descriptions
    - Data Requirements
    - External Interface Requirements
    - System Features
2. **Identify Assumptions and Constraints**
    - List the assumptions considered while developing the system.
    - Identify technical, operational, economic, legal, and time-related constraints affecting the project.
2. **Validate Requirement Completeness and Consistency**
    - Verify whether all stakeholder requirements have been addressed.
    - Check for ambiguity, redundancy, inconsistency, and missing requirements.
    - Suggest improvements to enhance the quality and completeness of the requirement specification.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modeling	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.

Assumptions, Constraints, and Requirement Validation	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	**Mark**
Scenario Understanding and Problem Analysis	**/5**
Functional and Non-Functional Requirement Identification	**/5**
Actor Identification and Use Case Modeling	**/5**
Software Requirement Specification (SRS) Documentation	**/5**
Assumptions, Constraints, and Requirement Validation	**/5**
TOTAL	**/5**

Study and Analyze the Scenario

1.1 Introduction

Education is an important right for every student. However, students with visual impairments may experience difficulties while using conventional online learning platforms. Many learning websites are not fully compatible with screen readers and may contain complex navigation, images without alternative text, small text, inaccessible learning materials, and difficult online assessments.

The proposed **Accessible Learning Platform** is designed to provide an inclusive digital learning environment. The platform supports:

- Screen reader compatibility
- Voice navigation
- Text-to-speech support
- Audio learning materials
- Keyboard navigation
- High-contrast mode
- Accessible online quizzes
- Student progress tracking

The system supports **United Nations Sustainable Development Goal 4 – Quality Education** by promoting inclusive and equitable education for students with visual impairments.

1.2 Business Domain

The business domain of this project is:

Education Technology (EdTech) and Accessible Digital Learning

The platform helps educational institutions provide accessible online education to students with visual impairments.

1.3 Problem Description

Students with visual impairments face several difficulties while using traditional online learning platforms. Some educational websites are designed mainly for students who can view and interact with visual content.

The main problems are:

- Limited screen reader support.
- Complex menus and navigation.

- Images without alternative text.
- Lack of voice-based navigation.
- Limited audio learning resources.
- Inaccessible online quizzes.
- Difficult mouse-based interaction.
- Limited support for keyboard navigation.
- Lack of personalized accessibility settings.
- Students may depend on teachers or other people to access learning content.

These problems reduce independent learning and may create unequal educational opportunities.

1.4 Existing Challenges

S.No.	Existing Challenge	Impact
1	Limited screen reader support	Students cannot access all content
2	Complex navigation	Difficult to move between pages
3	Lack of audio lessons	Learning depends mainly on visual text
4	Inaccessible quizzes	Students may not complete assessments independently
5	No voice navigation	Users depend on keyboards or other assistance
6	Images without alternative text	Important information may not be available
7	Limited accessibility settings	Users cannot personalize the platform
8	Lack of progress support	Students may find it difficult to monitor learning

1.5 System Objectives

The objectives of the proposed system are:

1. To develop an accessible online learning platform.
2. To support students with visual impairments.
3. To provide screen reader-compatible web pages.
4. To enable voice-based navigation.
5. To provide audio learning materials.

6. To support keyboard-based navigation.
7. To provide accessible quizzes and assessments.
8. To track student learning progress.
9. To help teachers upload accessible learning materials.
10. To enable administrators to manage users, courses, and reports.
11. To promote inclusive education under **SDG 4 – Quality Education**.

2. Scope of the System

2.1 In Scope

The following functions are included in the proposed system:

- Student registration and login.
- Teacher login.
- Administrator login.
- User profile management.
- Personalized accessibility settings.
- Screen reader support.
- Text-to-speech support.
- Voice navigation.
- Keyboard navigation.
- High-contrast mode.
- Course management.
- Audio learning lessons.
- Learning material upload.
- Accessible quizzes.
- Quiz evaluation.
- Student progress tracking.
- Performance reports.
- Accessibility feedback.
- Administrator dashboard.

2.2 Out of Scope

The following features are not included in the first version:

- Live video classes.
- Online payment and subscription.
- Social media integration.
- Virtual reality learning.
- Hardware-based assistive devices.
- Advanced artificial intelligence tutoring.
- Offline learning without downloaded content.
- Full multilingual support for all languages.

3. Functional Requirements

Functional requirements describe **what the system must do**.

ID	Functional Requirement
FR1	The system shall allow students, teachers, and administrators to register and log in.
FR2	The system shall authenticate users using valid login credentials.
FR3	The system shall provide role-based access for students, teachers, and administrators.
FR4	The system shall allow students to view available courses.
FR5	The system shall provide screen reader-compatible learning pages.
FR6	The system shall provide text-to-speech functionality for learning content.
FR7	The system shall allow users to control the platform using voice commands.
FR8	The system shall support keyboard navigation.
FR9	The system shall provide high-contrast mode and text-size controls.
FR10	The system shall allow teachers to create and manage courses.
FR11	The system shall allow teachers to upload learning materials and audio lessons.
FR12	The system shall allow students to access and listen to learning materials.
FR13	The system shall provide accessible quizzes and assessments.
FR14	The system shall evaluate quiz answers and display scores.

FR15	The system shall store student quiz results.
FR16	The system shall track completed lessons and learning progress.
FR17	The system shall display student performance reports.
FR18	The system shall allow students to submit accessibility feedback.
FR19	The system shall allow administrators to manage users.
FR20	The system shall allow administrators to generate reports.

4. Non-Functional Requirements

Non-functional requirements describe **how the system should perform**.

ID	Requirement	Description
NFR1	Performance	The system should respond to normal user requests within 3 seconds.
NFR2	Security	User passwords should be securely stored and protected.
NFR3	Reliability	The system should provide correct and consistent results.
NFR4	Usability	The interface should be simple and easy to use.
NFR5	Accessibility	The platform should support screen readers, keyboard navigation, and accessible labels.
NFR6	Availability	The system should be available whenever users need access.
NFR7	Scalability	The system should support an increasing number of users and courses.
NFR8	Maintainability	The software should be modular and easy to update.
NFR9	Portability	The platform should work on computers, tablets, and mobile devices.
NFR10	Compatibility	The system should work with modern web browsers.
NFR11	Data Privacy	User information should be accessed only by authorized users.
NFR12	Backup	Important user and learning data should be backed up regularly.

5. Actors and Use Cases

5.1 Primary Actors

Actor	**Role**
Student	Accesses courses, audio lessons, quizzes, and progress reports
Teacher	Creates courses, uploads learning materials, and manages quizzes
Administrator	Manages users, courses, reports, and system activities

5.2 Secondary Actors

Actor	**Role**
Screen Reader	Reads web content to visually impaired students
Voice Recognition Service	Receives and processes voice commands
Text-to-Speech Service	Converts learning text into audio
Database System	Stores user, course, quiz, and progress information

5.3 Major Use Cases

6. Use Case Descriptions

UC1 – Student Registration

Field	**Description**

Use Case ID	UC1
Use Case Name	Student Registration
Primary Actor	Student
Precondition	The student does not have an account
Main Flow	The student enters name, email, password, and accessibility preferences
System Response	The system validates the information and creates an account
Postcondition	A new student account is stored in the database

UC2 – User Login

Field	**Description**
Use Case ID	UC2
Use Case Name	User Login
Primary Actor	Student, Teacher, Administrator
Precondition	The user has a registered account
Main Flow	The user enters email and password
System Response	The system verifies the credentials
Postcondition	The user is redirected to the correct dashboard

UC3 – Voice Navigation

Field	**Description**
Use Case ID	UC3
Use Case Name	Voice Navigation
Primary Actor	Student
Precondition	The student is logged in and microphone access is enabled
Main Flow	The student gives a voice command such as “Open Courses”
System Response	The system recognizes the command and opens the requested page
Postcondition	The selected feature is displayed

UC4 – Audio Learning

Field	Description
Use Case ID	UC4
Use Case Name	Audio Learning
Primary Actor	Student
Precondition	The student is logged in and a lesson is available
Main Flow	The student opens a lesson and selects “Play Audio”
System Response	The system reads the learning content aloud
Postcondition	The student can listen to the lesson

UC5 – Accessible Quiz

Field	Description
Use Case ID	UC5
Use Case Name	Take Accessible Quiz
Primary Actor	Student
Precondition	The student has opened a course with an available quiz
Main Flow	The system displays or reads the questions and options
System Response	The student selects answers and submits the quiz
Postcondition	The system calculates and stores the score

UC6 – Upload Learning Content

Field	Description
Use Case ID	UC6
Use Case Name	Upload Learning Content
Primary Actor	Teacher
Precondition	The teacher is logged in
Main Flow	The teacher selects a course and uploads learning material
System Response	The system validates and stores the content

Postcondition	Students can access the published content

UC7 – Manage Users

Field	Description
Use Case ID	UC7
Use Case Name	Manage Users
Primary Actor	Administrator
Precondition	The administrator is logged in
Main Flow	The administrator views, updates, activates, or removes users
System Response	The system updates the user information
Postcondition	User records are updated

7. Data Requirements

The system requires the following data.

7.1 User Data

- User ID
- Name
- Email
- Password
- Role
- Account status
- Registration date

7.2 Accessibility Preference Data

- Preference ID
- User ID
- Voice speed
- Preferred language
- Font size
- High-contrast setting
- Navigation preference

7.3 Course Data

- Course ID
- Course name
- Course description
- Teacher ID
- Creation date

7.4 Lesson Data

- Lesson ID
- Course ID
- Lesson title
- Learning content
- Audio file
- Upload date

7.5 Quiz Data

- Quiz ID
- Course ID
- Question
- Option A
- Option B
- Option C
- Option D
- Correct answer

7.6 Progress Data

- Progress ID
- Student ID
- Course ID
- Completed lessons
- Quiz score
- Completion percentage

7.7 Feedback Data

- Feedback ID
- Student ID
- Feedback description
- Issue category
- Submission date
- Status

8. External Interface Requirements

8.1 User Interface Requirements

The user interface should provide:

- Clear headings.
- Proper labels for all input fields.
- Keyboard-accessible controls.
- High-contrast mode.
- Adjustable text size.
- Screen reader-compatible content.
- Audio controls.
- Voice navigation controls.
- Simple and consistent menus.

8.2 Hardware Interface Requirements

The system may require:

- Computer or laptop.
- Smartphone or tablet.
- Keyboard.
- Speaker or headphones.
- Microphone for voice navigation.
- Internet connection.

8.3 Software Interface Requirements

Software	Purpose
Python	Main programming language

Flask	Backend web framework
HTML	Web page structure
CSS	User interface design
JavaScript	Interactive and voice features
SQLite / MySQL	Database management
Web Speech API	Speech recognition and text-to-speech
Web Browser	Accessing the platform

8.4 Communication Interface

The system communicates using:

- HTTP and HTTPS.
- Web browser requests.
- Database connections.
- Voice service APIs.
- Text-to-speech services.

9. System Features

Feature 1 – User Management

The system provides:

- Registration.
- Login.
- Logout.
- User profile management.
- Role-based access.

Feature 2 – Personalized Accessibility

The system allows users to select:

- Voice speed.
- Text size.
- High-contrast mode.
- Preferred language.
- Navigation method.

Feature 3 – Voice Navigation

Students can use voice commands such as:

- “Open Courses”
- “Start Lesson”
- “Open Quiz”
- “Show Progress”
- “Go Home”

Feature 4 – Audio Learning

The system provides:

- Play audio.
- Pause audio.
- Repeat content.
- Change voice speed.
- Listen to learning materials.

Feature 5 – Course Management

Teachers can:

- Create courses.
- Update courses.
- Upload learning materials.
- Upload audio lessons.
- Publish learning content.

Feature 6 – Accessible Assessment

The system provides:

- Accessible questions.
- Screen reader support.
- Keyboard navigation.
- Audio reading of questions.
- Quiz evaluation.
- Score display.

Feature 7 – Progress Tracking

Students can view:

- Completed lessons.
- Quiz scores.
- Course completion percentage.
- Performance information.

Feature 8 – Administrator Dashboard

The administrator can:

- Manage users.
- Manage courses.
- View student activity.
- View accessibility feedback.
- Generate reports.

10. Assumptions

The following assumptions are considered:

1. Users have access to a computer or smartphone.
2. Users have an internet connection.
3. Students have basic knowledge of using a web application.
4. Users allow microphone access for voice navigation.
5. The browser supports the Web Speech API.
6. Teachers upload valid learning materials.
7. The educational institution provides authorized user access.
8. Users have speakers or headphones for audio learning.
9. The database server is available.
10. Administrators regularly maintain the system.

11. Constraints

11.1 Technical Constraints

- Voice recognition may not work in every browser.
- Speech recognition accuracy may depend on pronunciation and background noise.

- Internet connection is required for some voice services.
- Some older devices may have limited browser support.

11.2 Operational Constraints

- Users may require training to use voice commands.
- Teachers must prepare accessible learning materials.
- The system requires regular maintenance.

11.3 Economic Constraints

- The project may have a limited development budget.
- Advanced voice services may require paid APIs.
- Cloud hosting may create additional costs.

11.4 Legal and Accessibility Constraints

- User information must be protected.
- The system should follow accessibility guidelines.
- Educational content must follow copyright rules.
- User data should be accessed only by authorized users.

11.5 Time Constraints

- The project must be completed within the academic schedule.
- Testing time may be limited.
- Advanced features may be developed in future versions.

12. Requirement Validation

The requirements are validated using the following checks.

12.1 Completeness

The requirements cover:

- User registration.
- Login.
- Accessibility features.
- Courses.
- Audio lessons.
- Quizzes.

- Progress tracking.
- Teacher functions.
- Administrator functions.
- Database requirements.

Therefore, the major stakeholder requirements are included.

12.2 Consistency

The requirements do not conflict with each other.

For example:

- Students can access courses.
- Teachers can manage learning content.
- Administrators can manage users and reports.

Each user has a separate role and permission.

12.3 Clarity

Each requirement uses clear terms such as:

- "The system shall allow…"
- "The system shall provide…"
- "The system shall store…"

This reduces ambiguity.

12.4 No Redundancy

Similar requirements are grouped into modules.

For example:

- Voice navigation is included under the Accessibility Module.
- Course and lesson functions are included under the Learning Module.
- User management is included under the Administration Module.

12.5 Requirement Traceability

Stakeholder Need	Related Requirement

Students need accessible learning	FR5, FR6, FR7, FR8, FR9
Students need audio lessons	FR6 and FR12
Students need accessible assessments	FR13 and FR14
Teachers need content management	FR10 and FR11
Administrators need system control	FR19 and FR20
Institutions need student reports	FR16 and FR17

13. Suggested Improvements

The following improvements can be added in future versions:

- Mobile application.
- Multilingual voice support.
- AI-based learning assistant.
- Personalized learning recommendations.
- Live teacher support.
- Video lessons with audio descriptions.
- Offline learning support.
- Cloud deployment.
- Advanced accessibility analytics.
- Automatic accessibility checking for uploaded content.

14. Conclusion

The **Accessible Learning Platform** is designed to reduce accessibility barriers faced by students with visual impairments. The system provides screen reader support, voice navigation, audio learning, keyboard navigation, high-contrast mode, accessible assessments, and progress tracking.

The platform supports students, teachers, and administrators through separate modules and role-based access. The proposed system promotes independent learning and improves access to digital education.

This project contributes to **SDG 4 – Quality Education** by supporting inclusive, equitable, and accessible learning opportunities for all students.